

MAGGIE—Medical Application Game-Generator for Intelligence Evaluation

Development Plan

Bassel Taleb

Summer Higgins

Malek Harajli

Humayra Nawshin

Client: Dr. Stephen Farrow

GTA: Yermakhan Magzym

Wayne State University - Senior Capstone CSC4996

Dr. Seyed Ziae Mousavi Mojab

Spring/Summer 2026

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1. Project Overview

Many patients leave a medical appointment unsure of what their medications are for, how their conditions and prescriptions connect, or why taking them correctly matters. This gap contributes to a large and persistent public-health problem: medications that are not taken as prescribed are linked to tens of thousands of avoidable deaths and hundreds of billions of dollars in added healthcare spending each year. MAGGIE is a mobile learning application that helps close this gap by turning medication education into short, personalized quiz games. When a patient answers a question incorrectly, the app explains the missed concept and asks a follow-up question, so learning improves each round. MAGGIE then produces a short summary for the patient's physician that describes only what happened inside the game, which concepts were missed and which were reviewed, leaving all clinical interpretation to the physician. The application works entirely with simulated practice data from a medical-records test environment. Therefore, it does not use any real patient information, and it does not diagnose, prescribe, or give medical advice.

2. Project Purpose, Scope, and Objectives

2.1 Purpose

The purpose of MAGGIE is to support medication comprehension by turning medication education into an interactive, repeatable experience. Our client's research hypothesis is that iterative, game-based education can help patients better understand what their medications are for, how those medications connect to their health conditions, and where their misunderstandings remain. The project also reflects a core security requirement set by our client: Our team will develop and demonstrate the system using simulated sandbox data only, never real protected health information (PHI).

The physician-summary feature exists to preserve clinical boundaries. MAGGIE may summarize what happened inside the game, but it will not infer diagnoses, recommend treatment, change prescriptions, or tell a patient what to do medically. A physician can review the summary and draw any clinical conclusions independently.

2.2 Scope

In scope:

- A cross platform mobile prototype built with React Native and Expo for iOS and Android testing.
- Epic SMART on FHIR sandbox authorization and retrieval of simulated FHIR medication and diagnosis resources.
- A data mapping layer that converts sandbox FHIR resources into the internal game model.
- Personalized quiz generation from sandbox record data using client approved prompts, constraints, and educational wording.
- Adaptive remediation that explains missed concepts and asks a follow up question when the user answers incorrectly.
- Supabase storage for app side session records, quiz responses, scores, and remediation history.
- A physician-facing summary generated only from gameplay results, such as missed concepts, explanations shown, and session progress.
- Controlled demonstration and testing through Expo Go and the Supabase development project.

Out of scope:

- Use of real patient records, live hospital data, or any real protected health information.
- Public production deployment, app-store submission, or connection to a live EHR system.
- Diagnosis, treatment, prescribing, dosage adjustment, medication-adherence enforcement, or emergency guidance.

- Direct provider messaging, automatic transmission to a real physician, a physician portal, or EHR write back unless explicitly approved as a later scope change.
- Prescription refills, pharmacy ordering, payment processing, or insurance workflows.
- Independent medical-content research beyond sandbox data and any small, client-approved educational reference set.

2.3 Objectives

- By the end of Prototype 1, deliver an Expo mobile application skeleton connected to Supabase that can authorize against the Epic SMART on FHIR sandbox and retrieve simulated medication and diagnosis resources while demonstrating a secure data flow that uses no real PHI.
- By the end of Prototype 2, implement and display a complete quiz workflow using sandbox record data, including question generation, answer handling, scoring, and stored session results.
- By the end of the Final Product phase, implement adaptive remediation, follow-up questioning, persistent session history, physician-summary generation based only on gameplay activity, and a complete sandbox-only demonstration on iOS and Android through Expo Go. The final prototype will also include documentation and testing evidence showing that no real PHI is used, stored, or transmitted.

3. Team Organization

All team members contribute to implementation, testing, documentation, and presentations. The roles below identify primary ownership, so responsibilities remain clear.

Team Member	Primary Role	Key Responsibilities
Bassel Taleb	Team Lead / Technical Coordinator	Main point of contact for client, GTA, and instructor; runs meetings; coordinates sprint priorities; breaks tie votes; supports architecture decisions.
Summer Higgins	Documentation, QA Lead, Lead Notetaker	Owns written deliverables, formatting, proofreading, meeting notes, test case organization, and presentation materials, meeting minutes
Malek Harajli	Backend / Data Lead	Owns the Supabase schema, app side data storage, FHIR data parsing, API integration, and AI pipeline support.
Humayra Nawshin	Frontend / UI Lead	Owns Figma screens, the React Native interface, navigation, quiz screens, and patient facing usability improvements.

4. Problem Resolution Policies

4.1 Technical Disputes

For technical disagreements, the affected area lead will summarize the issue and explain how each proposed solution may affect implementation, project requirements, security, and privacy. Our team will review the information together and work toward consensus. If consensus cannot be reached, then our Team Lead will make the final decision after considering the group's input. Major technical decisions will be recorded in the meeting minutes, project documentation, or GitHub Issues so the reasoning remains traceable.

4.2 Security, Privacy, or Medical Questions

When a question involves patient data, deployment, medication information, or clinical interpretation, our team applies the safest interpretation of the user's condition until our client and GTA clarify the issue. The default policy is no real PHI, no public deployment, no medical advice, and no direct provider transmission unless approved in writing as a change of scope.

4.3 Team Member Issues

If a team member misses a commitment or becomes unresponsive, our team Lead checks in privately within 24 hours. If the issue remains unresolved after another 24 hours, our team meets to redistribute work and set a recovery plan. Continued issues are escalated to the GTA and, if necessary, the instructor.

4.4 Scope Changes and Schedule Non-adherence

Our team will not add or remove major features without documenting the change in the meeting minutes and in GitHub Issues. If a prototype milestone is at risk, our Team Lead must raise the issue at the next meeting and propose either a scope reduction for that prototype or a revised plan. Our team will reduce feature scope before reducing security, testing, or documentation quality.

5. Project Plan and Schedule

5.1 Meeting Cadence

- **Client / GTA meeting:** Online Zoom meeting on every Thursdays at 4:00 PM with Dr. Farrow and Yermakhan Magzym for feedback, demos, and clarification.
- **Internal team meeting:** Friday at 1:00 PM online for at least one hour, continuing longer when needed to resolve issues and/or assign tasks.
- **Asynchronous coordination:** WhatsApp for quick updates, GitHub Issues for task status, and a shared OneDrive for formal documents and meeting minutes.

5.2 Prototype Phases

Phase	Main Goal	Planned Deliverables
Prototype 1 (Weeks 1–4)	Foundational system and data pipeline	Expo / React Native project skeleton; Supabase development schema; Epic SMART on FHIR sandbox authorization; retrieval of simulated medication and diagnosis resources; first secure data-flow demo.
Prototype 2 (Weeks 5–9)	Core gameplay implementation	FHIR-to-game mapping; AI / prompt pipeline; quiz screen; answer handling; scoring; stored session results; initial UI review with client.
Final Product Phase (Weeks 10-12)	Adaptive remediation and reporting	Remediation after incorrect answers; follow up questioning; persistent remediation history; physician summary output; usability and accessibility refinements.
Final Product Phase (Weeks 10-12)	Stabilization and demo preparation	End to end sandbox demo on iOS and Android through Expo Go; bug fixing; testing evidence; final documentation; no new features unless required to fix a blocker.

6. Configuration Management Plan

6.1 Source Control and Branching

Our team uses a private GitHub repository and adds the required course and GTA accounts for progress checks. The main branch must always remain stable and demonstrable. A ‘develop’ branch is used for integration testing and where all completed features will go. New work is completed on its own specific feature branch (e.g., “feature/authentication”), and fixes will use their own branch as well. All merges into ‘develop’ or ‘main’ require a pull request and at least one review from another teammate.

6.2 Code Quality and Secrets

The codebase uses TypeScript, ESLint, and Prettier to keep code readable and consistent. Environment variables store Supabase keys, Epic sandbox configuration, and AI-service keys. Secrets, access tokens, exported data, and any local test files that could contain sensitive information must never be committed. The application will only use simulated sandbox data during development and demonstration.

6.3 Task Tracking and Documentation

GitHub Issues and a GitHub Projects board track the backlog, in-progress work, review, and completed tasks. Every pull request should reference the issue it closes. Meeting notes, client feedback, and major technical decisions are stored in the shared project folder and linked from the repository when relevant.

7. Technologies

Layer	Technology	Reason for Choice
Design	Figma & LibreSprite	Supports collaborative wireframing before implementation and gives our client a concrete way to review the patient flow,
Mobile Frontend	React Native + Expo + TypeScript	One codebase supports iOS and Android testing. Expo Go reduces native setup time, and TypeScript helps catch errors early.
Health Data Access	Epic SMART on FHIR + Epic FHIR Sandbox	Matches our client's EHR integration goal while keeping development limited to simulated test records.
Backend / Database	Supabase	Chosen as the single backend platform. PostgreSQL fits structured session, quiz, remediation, and summary records, and row level security supports safer access rules. Compared with Firebase, Supabase better matches relational reporting and avoids adding a second backend surface.
AI / Game Generation	OpenRouter API with client approved prompts	Allows model flexibility while keeping the generated quiz and remediation layer separate from clinical decision making. Outputs are constrained to educational gameplay content.
Medical-Content Boundary	Sandbox FHIR data + client approved educational wording	Medication questions are based on sandbox resources and approved prompt and content constraints. Our team will not independently publish unvalidated medical guidance.
Source Control	Git + GitHub	Supports traceable commits, pull requests, reviews, issues, project boards, and course/GTA progress checks.
Demo Environment	Expo Go + Supabase development project	Supports a controlled demonstration without public deployment, app store release, real PHI, or live EHR integration.

8. Summary

Our development plan defines a focused workflow for building MAGGIE within one semester. The plan preserves our client's core idea, which is a smartphone-based medication learning game that uses EHR style data and iterative remediation, while narrowing implementation to a safe capstone scope: Epic sandbox data, Supabase app side storage, controlled demonstration, and a physician summary based only on game activity. If later client feedback changes the scope, then our team will document the reason, update the plan, and adjust the prototype schedule accordingly.